Premature Accountability Convergence in Regulated Organizations

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Abstract Responsibility in regulated organizations is not a singular event but a temporally structured accomplishment — anticipatorily aligned before action, procedurally embedded during execution, and retrospectively stabilized under closure pressure. As artificial intelligence systems become integrated into documentation and evaluative infrastructures, this pre-existing convergence dynamic does not disappear; it accelerates. This paper introduces **Premature Accountability Convergence (PAC)** to describe a conditional acceleration dynamic within the responsibility lifecycle. PAC occurs when representational compression, epistemic authority amplification, and temporal contraction combine to reduce interpretive variance before causal ambiguity has been meaningfully examined. Document analysis of FDA Warning Letters ($N = 50$, 160 violations, 2016–2025) establishes that responsibility convergence is a structural property of accountability documentation prior to any computational mediation: attribution language varies substantially across violations, yet final accountability converges onto the regulated firm in 159 of 160 cases. A two-study empirical program ($N = 20$ each) demonstrates both the behavioral baseline — early convergence reflecting institutional pressure rather than epistemic certainty — and that counter-convergence design can delay premature closure without automating conclusions. By conceptualizing AI as a representational accelerator rather than an autonomous decision-maker, PAC shifts attention from algorithmic bias toward the temporal architecture of accountability.

Keywords: premature accountability convergence, artificial intelligence, responsibility lifecycle, representational compression, organizational investigations, regulated environments, pharmaceutical manufacturing

Introduction

In 159 of 160 FDA Warning Letter violations issued to pharmaceutical manufacturers between 2016 and 2025, final accountability converges onto the regulated firm — regardless of whether the violation is described as an actor's "failure to" act or an "inadequate" organizational system. This pattern is not a product of computational mediation. It reflects a more fundamental dynamic: in regulated organizations, documentary form has long compressed interpretive space and concentrated accountability onto identifiable actors before any AI system enters the sensemaking process.

Yet as artificial intelligence systems become embedded within documentation, evaluation, and compliance infrastructures, this pre-existing convergence dynamic does not disappear — it accelerates. AI systems generate structured representations — summaries, classifications, anomaly flags, predictive indicators — that enter directly into accountability workflows. While frequently described as decision-support tools, these systems operate at a deeper infrastructural level: they intervene in the representational architecture through which responsibility becomes legible, assessable, and administratively actionable.

This pattern establishes a critical baseline: responsibility convergence is a structural property of accountability documentation in regulated environments, operating independently of computational mediation. What AI systems introduce is not convergence itself, but acceleration — a reshaping of the temporal dynamics through which interpretive space narrows and responsibility stabilizes.

Responsibility in regulated organizations is not a singular event but a temporally structured accomplishment. It is anticipatorily aligned before action, procedurally embedded during execution, and retrospectively stabilized under conditions of ambiguity and closure pressure. Prior work within this responsibility program has conceptualized how expectations converge prospectively (Kim, 2026a) and how attribution stabilizes retrospectively through investigative reasoning (Kim, 2026c). These frameworks explain how responsibility becomes structured and consolidated across stages of organizational life.

What remains theoretically underspecified, however, is the temporal architecture of convergence itself. Existing theories clarify how responsibility aligns and stabilizes; they do not fully account for how infrastructural mediation may alter the tempo and breadth of that stabilization. As AI systems become embedded within documentation and evaluative processes, interpretive space may narrow earlier in the sensemaking sequence. Convergence may occur more rapidly — not because actors intend closure, nor because institutions become more rigid, but because computational mediation compresses representational diversity and accelerates inscription.

Without a framework that isolates this acceleration dynamic, the effects of AI on accountability risk being misattributed to bias, individual cognition, or institutional design alone. The missing dimension is temporal: how quickly responsibility stabilizes, how much interpretive variance persists before closure, and how infrastructural mediation reshapes these dynamics.

This paper introduces **Premature Accountability Convergence (PAC)** to conceptualize this acceleration mechanism. PAC identifies the conditions under which computational mediation compresses interpretive space and hastens responsibility stabilization within the responsibility lifecycle. Rather than asking whether AI changes who is responsible, this framework asks how AI reshapes the timing, durability, and scope of responsibility convergence. In doing so, it advances a temporal turn in the theory of organizational accountability.

The remainder of this paper is organized as follows. Section 2 develops the responsibility

lifecycle as a theoretical foundation. Section 3 situates computational mediation within this lifecycle. Section 4 introduces PAC and its three mechanisms. Section 5 presents theoretical propositions. Section 6 provides the empirical illustration from FDA Warning Letter analysis. Section 7 discusses theoretical contributions and implications. Section 8 concludes.

Responsibility as a Multi-Stage Organizational Lifecycle

Responsibility in regulated sociotechnical organizations unfolds across temporally distinct yet structurally interconnected stages. Rather than emerging solely at the moment of investigation or formal evaluation, responsibility is progressively produced through anticipatory alignment, procedural embedding, and retrospective stabilization (Perrow, 1984; Vaughan, 1996). These stages together constitute a lifecycle through which accountability expectations are generated, operationalized, and ultimately consolidated into administratively actionable forms.

Crucially, responsibility convergence does not occur at a single point. It accumulates across stages. Interpretive variance narrows prospectively, is structured evidentially during execution, and is formally condensed under retrospective closure pressure. Understanding this lifecycle clarifies how accountability is temporally organized prior to the introduction of computational mediation.

Anticipatory Structuring of Responsibility

In regulated environments, organizational actors rarely act without anticipation of future evaluation. Accountability expectations operate prospectively, shaping how obligations, risks, and acceptable conduct are interpreted before execution. The possibility of inspection, audit, or review functions as a structuring condition of action rather than a subsequent add-on.

Anticipatory alignment occurs when actors orient behavior toward expected evaluative standards. Responsibility is therefore structured ex ante: individuals interpret instructions, exercise discretion, and assess risk with awareness of how these actions may later be scrutinized. In pharmaceutical GMP environments, quality control analysts document interpretive decisions not merely to record outcomes but to produce accounts that will withstand regulatory inspection — orienting behavior toward anticipated auditability before any deviation or investigation occurs (Power, 1997). At this stage, convergence operates prospectively: interpretive variance narrows not through formal attribution, but through alignment with anticipated evaluative criteria.

Procedural Embedding During Execution

During execution, responsibility expectations become embedded within documentation systems, reporting architectures, and procedural infrastructures. Standard operating procedures,

logging requirements, validation protocols, and compliance frameworks structure how events are recorded and translated into organizational memory.

Procedural embedding stabilizes responsibility not by assigning blame, but by structuring the evidentiary pathways through which future attribution will unfold. Documentation formats define what counts as relevant information, how deviations are categorized, and which explanatory variables become visible within formal records. In doing so, procedural infrastructures pre-format the representational space within which retrospective interpretation will occur. Convergence at this stage remains incomplete but accumulative: interpretive possibilities are not eliminated, but channeled into administratively legible categories.

Retrospective Stabilization of Responsibility

After outcomes occur, responsibility is articulated and consolidated through investigations, reports, evaluative documentation, and corrective action processes. Under conditions of causal ambiguity, organizations must translate distributed events into administratively actionable accounts. This requires selecting attributional anchors that can stabilize accountability within formal structures.

Investigative reasoning frequently unfolds under closure pressure: ambiguous causal pathways must be rendered determinate in order to justify corrective action, compliance responses, or performance evaluation (Dekker, 2007; Vaughan, 1996). Through this process, interpretive alternatives are gradually reduced as coherent explanatory narratives acquire institutional authority. The FDA Warning Letter data illustrates this dynamic at the regulatory level: across violation categories, final attribution converges onto the regulated firm regardless of whether the evidentiary basis supports actor-centered or system-centered causal accounts.

Retrospective stabilization marks the formal consolidation of responsibility convergence. Ambiguity is condensed into administratively durable accounts that become embedded in documentation systems and organizational memory. At this stage, responsibility is no longer anticipatory but stabilized; convergence is driven by the need for closure and institutional legibility (Weick, 1995).

Taken together, these stages illustrate that responsibility convergence is not a discrete event but a cumulative and temporally distributed organizational process. Each stage narrows interpretive variance, structures evidentiary pathways, and consolidates accountability in ways that shape the conditions under which subsequent stages unfold.

Computational Mediation within the Responsibility Lifecycle

Having established responsibility as a temporally structured organizational lifecycle, we now consider how computational systems enter and interact with this process. Artificial intelligence systems are increasingly embedded within procedural workflows, documentation regimes, and evaluative infrastructures across regulated domains. Rather than functioning as autonomous decision-makers, these systems operate as representational mediators that translate organizational events into structured, recordable, and evaluable forms.

Computational outputs — summaries, classifications, anomaly flags, pattern detections, and predictive indicators — do not independently assign responsibility. However, they shape how events are rendered legible within organizational documentation and evaluative processes. In pharmaceutical quality systems, for instance, electronic quality management platforms structure deviation investigation workflows through mandatory completion fields, approval routing, and classification schemas — converting open-ended causal inquiry into sequenced administrative processes with predefined closure criteria. The representational architecture of these systems does not determine investigative conclusions, but it shapes which causal pathways remain visible and administratively actionable within formal documentation.

Importantly, computational mediation does not introduce responsibility structures ex nihilo. The anticipatory alignment, procedural embedding, and retrospective stabilization of responsibility precede the integration of AI systems. As the FDA Warning Letter analysis demonstrates, organizational infrastructures already generate expectations of accountability and stabilize attribution through documentary form — independently of computational mediation. What computational mediation alters is not the existence of responsibility convergence but its tempo and representational form.

Because algorithmically generated representations often carry the perceived authority of formalized calculation, they may narrow interpretive divergence earlier in the sensemaking process. Structured outputs can foreground particular causal pathways while backgrounding others, shaping which interpretations appear administratively actionable. In doing so, computational systems may compress the temporal space between ambiguity and stabilization within the responsibility lifecycle. Understanding this acceleration dynamic requires a framework that specifies the conditions under which computational mediation intensifies convergence.

Premature Accountability Convergence (PAC)

Computational mediation does not introduce responsibility structures into organizational systems; rather, it interacts with a pre-existing lifecycle of convergence. Within this lifecycle, AI systems function as representational accelerators. They do not determine who is responsible,

but they alter the tempo and breadth through which responsibility stabilizes.

Premature Accountability Convergence (PAC) identifies a structural acceleration tendency that becomes visible under conditions of computational mediation. PAC occurs when representational compression, perceived epistemic authority, and temporal contraction combine to reduce interpretive variance before causal ambiguity has been meaningfully examined. The lifecycle itself remains intact; what shifts is the timing and scope of stabilization.

PAC is not a claim that AI inherently distorts accountability. Rather, it specifies a conditional infrastructural dynamic: when structured computational outputs enter documentation and evaluative systems early in the sensemaking sequence, responsibility may converge more rapidly and with diminished interpretive plurality. PAC therefore reframes computational mediation as a temporal intervention within accountability architecture — drawing attention to how infrastructural design shapes the durability of ambiguity, the persistence of alternative explanations, and the pace at which attribution becomes administratively actionable.

Three interrelated mechanisms contribute to PAC.

Representational Compression

Computational systems translate complex and distributed organizational events into structured outputs — summaries, classifications, risk scores, anomaly flags, and predictive indicators. In doing so, they compress interpretive multiplicity into standardized representational forms. This compression enhances administrative clarity and traceability, but it also pre-selects which explanatory variables remain visible within formal documentation. As events are formatted into predefined categories, the range of plausible causal accounts may narrow prior to full deliberation.

The FDA Warning Letter corpus illustrates how representational compression operates even prior to computational mediation. Across six violation clusters, attribution language varies substantially in causal framing, yet the documentary form of the Warning Letter compresses this variation into uniform firm-level accountability in 159 of 160 cases. The representational architecture of regulatory documentation pre-selects the attributional anchor before any investigative deliberation occurs. Computational systems intensify this dynamic by further standardizing the categories through which organizational events are rendered legible — accelerating the compression that documentary form already initiates.

Epistemic Authority Amplification

Algorithmically generated outputs often carry perceived legitimacy derived from formalized calculation, technical infrastructure, and data-driven framing. When computational representations are treated as epistemically authoritative, interpretive contestation may diminish. Epistemic au-

thority amplification does not require blind trust in algorithms; rather, it emerges when structured outputs become embedded within formal documentation systems and are granted institutional weight.

In regulated environments, this dynamic is observable when electronic quality management systems generate automated deviation classifications or risk scores that become the starting point for investigative reasoning rather than one input among many. Alternative systemic or structural explanations may receive diminished scrutiny as a result, even when evidentiary ambiguity would otherwise support their consideration.

Temporal Contraction of Sensemaking

Computational mediation frequently reduces the interval between event occurrence and formal inscription. Automated logging systems, real-time dashboards, and immediate classification tools shorten the deliberative window within which alternative interpretations may circulate informally. When documentation is generated rapidly and embedded early within evaluative infrastructures, preliminary interpretations acquire durability before distributed explanations are fully articulated.

In pharmaceutical GMP environments, deviation management systems impose mandatory investigation closure timelines — typically 30 days for minor deviations. When AI-assisted classification tools generate preliminary root cause categories at the point of deviation initiation, those categories acquire durability within the investigation record before the full evidentiary picture has emerged. The deliberative interval within which alternative interpretations might circulate contracts around computationally inscribed preliminary accounts.

Representational compression constitutes the primary structural driver of PAC. Its intensity depends on contextual amplification mechanisms — the perceived epistemic authority granted to computational outputs and the temporal interval available for interpretive deliberation. When these conditions converge, responsibility stabilization may occur earlier and with diminished interpretive plurality.

Theoretical Propositions

The PAC framework yields four theoretical propositions specifying how variation in representational compression, epistemic authority, and temporal contraction conditions the strength and distribution of convergence across organizational contexts.

Proposition 1: Representational Compression

The greater the degree of computational representational compression within organizational documentation systems, the more likely interpretive space will narrow prior to full causal exploration.

Representational compression refers to the extent to which complex, distributed, and ambiguous events are translated into standardized, categorical, or summarized formats. While such structuring enhances administrative clarity and efficiency, it simultaneously limits the range of causal alternatives that remain visible within formal documentation. This proposition is non-obvious in an important respect: representational compression does not require intentional simplification or algorithmic bias. It operates structurally — through the design of documentation formats, classification schemas, and reporting architectures — regardless of the intentions of system designers or investigators. The FDA Warning Letter corpus illustrates this at the pre-computational level: documentary form produces uniform firm-level attribution across causally heterogeneous violations not because investigators intend to simplify, but because the representational architecture of the document is organized to produce administratively actionable attribution.

Proposition 2: Epistemic Authority

The stronger the perceived epistemic authority of computational outputs, the more rapidly responsibility expectations will align around computationally foregrounded interpretations.

Because computational representations derive legitimacy from formalized calculation and technical infrastructure, actors may defer to their outputs with reduced contestation (Porter, 1995). In contexts of ambiguity, perceived data-driven precision can accelerate interpretive alignment by stabilizing particular explanatory frames early in the sensemaking process. What makes this proposition theoretically significant is the asymmetry it introduces: epistemic authority amplification does not require that computational outputs be accurate or complete — it requires only that they be institutionally credible. In regulated environments where documentation systems carry audit authority, computationally generated classifications may acquire institutional weight independent of their evidentiary completeness.

Proposition 3: Temporal Contraction

The shorter the temporal interval between event occurrence and formal computational documentation, the greater the likelihood of premature responsibility stabilization.

Temporal contraction of sensemaking refers to the reduction in deliberative interval between an event and its formal inscription within documentation systems. When documentation is

generated rapidly and becomes embedded within evaluative infrastructures, early interpretations gain durability and institutional visibility. As a result, responsibility accounts may solidify before distributed or systemic explanations are meaningfully articulated. This proposition connects PAC to broader organizational theories of sensemaking and closure pressure (Weick, 1995): temporal contraction does not eliminate interpretive diversity — it forecloses it institutionally by granting durability to preliminary accounts before alternatives are fully examined.

Proposition 4: Counter-Convergence Design

Deliberate design interventions that expand interpretive time, distribute epistemic authority, and preserve representational plurality can delay or mitigate premature accountability convergence.

PAC is not an inevitable consequence of computational mediation. Its intensity depends on infrastructural conditions that are design variables. Investigative systems that require explicit comparison across multiple causal hypotheses before closure, that treat computational classifications as provisional inputs rather than authoritative determinations, and that preserve temporal buffering between event inscription and formal attribution can reduce the acceleration dynamic identified by PAC. This proposition is grounded in the scaffolded intervention component of this study, detailed in Section 6.

Empirical Illustration: Responsibility Convergence in FDA Regulatory Documentation

The theoretical framework specifies PAC as a conditional acceleration dynamic within the responsibility lifecycle. Before examining how computational mediation intensifies this dynamic, it is necessary to establish its structural baseline: the extent to which responsibility convergence already operates within regulatory documentation prior to any computational intervention. Analysis of FDA Warning Letters issued to finished pharmaceutical manufacturers provides this empirical grounding, supplemented by a two-study empirical program examining convergence at the investigative level.

Data and Method

The corpus consists of 50 FDA Warning Letters issued to finished pharmaceutical drug product manufacturers between 2016 and 2025, sampled at five cases per year to ensure temporal coverage. The unit of analysis is the violation block within each Warning Letter, yielding 160 violation observations. Each violation block was coded along two dimensions using a rule-based computational coding procedure applied consistently across the full corpus. Attribution language

was classified as action-oriented ("failure to"), state-oriented ("inadequate"), or residual. Violation type was classified into six clusters: investigation failures, oversight absence, validation deficiencies, data integrity lapses, release and stability issues, and equipment and environmental conditions. The coding scripts, criteria definitions, and violation-level dataset are publicly available at `https://github.com/sorakim-lab/fda-warning-letters-pac-analysis`.

Participants in both empirical studies were recruited from within the author's pharmaceutical GMP team, all with direct experience in deviation investigation or CAPA processes.

Findings: Documentary Convergence

Across 160 violation observations, attribution language distributes as follows: action-oriented language in 39 violations (24.4%), state-oriented language in 106 violations (66.2%), and residual formulations in 15 violations (9.4%). Despite this variation in attributional framing, final accountability converges uniformly onto the regulated firm: 159 of 160 violations (99.4%) attribute responsibility to the Firm. A single violation (0.6%) attributes responsibility to the Quality Control Unit as a distinct subunit. No violation attributes responsibility to individual actors, distributed systemic conditions, or external factors.

This convergence is structurally independent of attribution language. Violations framed in action-oriented terms converge onto the Firm at the same rate as violations framed in state-oriented terms. Documentary form — the Warning Letter as a regulatory instrument — compresses causally heterogeneous violation descriptions into a uniform attributional outcome. The representational architecture of the document produces administratively actionable attribution regardless of the causal account embedded within individual violation blocks.

Examination of attribution language across violation clusters reveals systematic variation in framing while final accountability remains uniformly convergent. Investigation failures (cluster 1, $n = 18$) exhibit the highest concentration of action-oriented language (67%), foregrounding identifiable procedural conduct. Validation deficiency violations (cluster 3, $n = 15$) exhibit exclusively state-oriented language (100%), describing systemic states — yet produce identical firm-level attribution. Equipment and environmental violations (cluster 6, $n = 38$) similarly exhibit high concentrations of state-oriented language (89%) while maintaining uniform firm-level convergence. This cluster pattern most clearly illustrates representational compression: violations entirely framed in systemic terms nonetheless produce actor-centered attribution as their documentary outcome.

Findings: Investigative Convergence and Counter-Convergence Design

A pilot study ($N = 20$) established the behavioral baseline for convergence under closure pressure. Across participants with direct GMP investigation experience, 15 of 20 produced single-cause explanations at the initial timepoint. Confidence ratings averaged 1 out of 5 regardless of experience level — indicating that early convergence reflects institutional pressure rather than epistemic certainty. This finding establishes that premature convergence is not a product of investigator inadequacy but of the structural conditions under which investigative reasoning unfolds.

A scaffolded AI intervention ($N = 20$) demonstrated that counter-convergence design can delay premature closure. With structured reasoning support, 18 of 20 participants broadened causal reasoning beyond their initial single-cause explanation. 15 of 20 reconsidered their initial attribution, with 12 of those 15 shifting from person-centered to systemic or procedural explanations. Usability ratings averaged 5.0 out of 5 and all 20 participants expressed adoption intent. The primary adoption barrier was conceptual rather than interactional: the interface was perceived as clear, but the notion that investigative reasoning deserves structured support was unfamiliar in GMP contexts. Replication at scale remains necessary.

Theoretical Implications

These findings establish three empirical observations that ground the PAC framework. First, responsibility convergence is a structural property of accountability documentation in regulated environments, operating prior to and independently of computational mediation. Second, representational compression operates at the level of documentary architecture rather than individual investigative reasoning — the convergence pattern across 160 violation observations reflects the structural organization of regulatory documentary form, not investigator intent or cognitive bias. Third, premature convergence at the investigative level reflects institutional pressure rather than epistemic certainty, and is amenable to design intervention. Together, these observations establish the empirical baseline against which PAC's acceleration dynamic must be understood.

Discussion

PAC contributes to sociotechnical and organizational theories of accountability by introducing acceleration as a critical analytical dimension within responsibility structuring. Prior research has examined how responsibility is anticipatorily aligned before action and how attribution stabilizes after events through investigative processes. Less attention has been paid to how computational infrastructures alter the tempo and breadth of stabilization across these stages. By

conceptualizing AI as a representational accelerator rather than an autonomous decision-maker, PAC extends accountability theory beyond attribution and compliance toward the temporal dynamics of responsibility production.

The FDA Warning Letter analysis establishes that responsibility convergence is not an artifact of computational mediation. Across 160 violation observations, documentary form produces uniform firm-level attribution regardless of causal framing, evidentiary complexity, or violation type. This baseline finding reframes the theoretical puzzle: the question is not whether AI introduces convergence into organizational systems, but how it reshapes the tempo, breadth, and durability of convergence dynamics that already exist. PAC answers this question by specifying the mechanisms through which computational mediation intensifies pre-existing stabilization tendencies.

The central tension exposed by PAC is not between automation and human judgment, but between speed and deliberation. As computational systems enhance administrative efficiency, they may simultaneously reduce the durability of ambiguity within organizational processes. Accountability systems differ not only in who they assign responsibility to, but in how long interpretive variance is permitted to endure before stabilization becomes institutionally binding. When interpretive space contracts too rapidly, distributed or structural explanations may lose durability before being meaningfully examined.

The scaffolded intervention findings extend this analysis toward design. The finding that 15 of 20 participants produced single-cause explanations with confidence averaging just 1 out of 5 establishes that premature convergence is not a product of investigator certainty but of institutional pressure. Counter-convergence design demonstrated that AI deployed to expand interpretive space — surfacing alternative causal accounts rather than foregrounding single attributions — represents a fundamentally different design orientation than AI deployed as an efficiency accelerator. The design challenge is not improving the accuracy of computational outputs but preserving the conditions under which human judgment remains meaningfully comparative.

Situating PAC within the responsibility program clarifies its relationship to adjacent frameworks. Anticipated accountability convergence (AAC) (Kim, 2026a) explains how expectations align before action through procedural and documentary infrastructures. Cognitive friction (CF) (Kim, 2026b) identifies how procedural texts embed localized judgment demands during execution. Responsibility convergence (RC) (Kim, 2026c) explains how attribution stabilizes after events under investigative closure pressure. PAC identifies how computational mediation may accelerate stabilization dynamics across these stages — and how deliberate counter-convergence design can resist that acceleration. Together, these frameworks trace responsibility structuring across the full temporal arc of regulated organizational work.

Accountability should therefore be understood not merely as a normative principle but as

an architecture of temporal structuring and evidentiary design. When computational systems compress interpretive space too rapidly, organizations may inadvertently reduce the durability of distributed explanations — producing accountability outcomes that are administratively efficient but causally impoverished. Preserving interpretive plurality requires deliberate design interventions: mechanisms for contestability, temporal buffering, layered documentation that preserves alternative interpretations, and procedural conditions under which stabilized accounts can be reopened.

Conclusion

This paper introduced Premature Accountability Convergence (PAC) as a conditional acceleration dynamic within contemporary accountability infrastructures. Rather than portraying AI as a determinant of responsibility, the framework conceptualized computational systems as representational mediators that compress interpretive space and intensify stabilization dynamics within the responsibility lifecycle.

The central contribution is not simply to observe that AI accelerates accountability processes, but to specify the structural mechanisms through which this acceleration operates, the empirical baseline against which it must be understood, and the design conditions under which it can be resisted. Document analysis of FDA Warning Letters establishes that responsibility convergence is a structural property of accountability documentation prior to any computational mediation: documentary form produces uniform attribution outcomes across causally heterogeneous violation descriptions, compressing interpretive diversity into administratively actionable accountability assignments regardless of evidentiary framing. The two-study empirical program demonstrates that premature convergence at the investigative level reflects institutional pressure rather than epistemic certainty, and that counter-convergence design can delay premature closure — though replication at scale remains necessary.

The three mechanisms identified — representational compression, epistemic authority amplification, and temporal contraction of sensemaking — specify how computational mediation intensifies pre-existing stabilization tendencies within the responsibility lifecycle. Organizations that deploy AI systems within investigative and evaluative workflows without attending to these mechanisms may inadvertently accelerate convergence toward actor-centered attribution at the expense of systemic causal reconstruction.

Situated within a broader responsibility lifecycle, PAC specifies the AI-mediated stage at which pre-existing convergence dynamics are accelerated through computational representation — following anticipatory alignment (AAC), procedural embedding (CF), and investigative stabilization (RC). Together, these frameworks trace responsibility structuring across the full temporal arc of regulated organizational work, from the prospective alignment of accountability

expectations through the procedural embedding of judgment demands, the investigative stabilization of attribution, and the representational acceleration of convergence through computational mediation. PAC thus completes the responsibility lifecycle by specifying how computational infrastructures reshape the temporal architecture of accountability in contemporary sociotechnical organizations.

Data Availability

FDA Warning Letters corpus: The 50 Warning Letters analyzed in this study are publicly available through the FDA's official database at `https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/compliance-actions-and-activities/warning-letters`.

The violation-level coding dataset ($N = 160$) is publicly available at `https://github.com/sorakim-lab/fda-warning-letters-pac-analysis`.

Empirical study data: Data from the two-study empirical program ($N = 20$ each) consist of participant reasoning protocols collected in pharmaceutical investigation contexts. Due to participant confidentiality, these data are not publicly available but may be obtained from the author upon reasonable request at sorakim.lab@gmail.com.

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